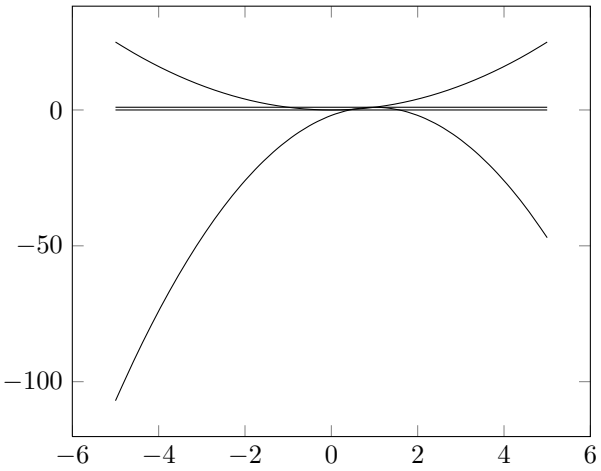$$F_x(x) = \begin{cases} 0 & \text{for } x < 0 \\ x^2 & 0 \le x < 0.5 \\ 1 - 3(1-x)^2 & 0.5 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$